

## Chapter 3

## Chapter 3: PIC 18 Microcontroller Architecture & Features

- 3.1 PIC Family Overview
- 3.2 Comparison of 8051 & PIC18
- 3.3 Features of PIC 18F458
- 3.4 Architecture of PIC 18F458
- 3.5 Memory organization
  - 3.5.1 ROM Organization
  - 3.5.2 Stack Memory
  - 3.5.3 RAM Organization
- 3.6 Instruction Cycle and pipelining
- 3.7 Clock Configuration

### 3.1 PIC Family Overview

PIC is first represented Peripheral Interface Controller, later it is expanded and is now known as Programmable Intelligent Computer. PIC is a family of microcontrollers and Microchip Technology Inc., USA manufactures them.

**Table 3.1: PIC Families**

|                              | <b>Baseline Architecture</b> | <b>Mid-Range Architecture</b>                     | <b>Enhanced Mid-Range Architecture</b>              |
|------------------------------|------------------------------|---------------------------------------------------|-----------------------------------------------------|
| <b>Performance</b>           | 5 MIPS                       | Upto 8 MIPS                                       | Upto 16 MIPS                                        |
| <b>Instruction Code Size</b> | 12-Bit                       | 14-Bit                                            | 16-Bit                                              |
| <b>Instructions</b>          | 33                           | Typical 35, upto 50 in Improved versions          | Typical 75, upto 83 in Improved versions            |
| <b>Data Memory</b>           | Upto 138 Bytes               | Upto 1.5 KB                                       | Upto 4KB                                            |
| <b>Program Memory</b>        | Upto 3 KB                    | Upto 28 KB                                        | Upto 128 KB                                         |
| <b>Hardware Stack</b>        | 2 Level                      | 8 to 16 Level                                     | 32 Level                                            |
| <b>Interrupts</b>            | No                           | Single interrupt capability with H/W context save | Multiple Interrupt capability with W/W context save |
| <b>Families</b>              | PIC10, PIC12, PIC16          | PIC12, PIC16                                      | PIC18                                               |

Microchip Technology Inc. also produces various microprocessors, Digital Signal Processors and various other devices. It also provides the required programming and development tools like ICD, MPLAB X IDE and various other tools.

There are different families of PIC Microcontrollers and are based on the features they provide. They are listed below. A basic comparison of them is presented in Table 3.1.

- Baseline architectures (PIC 10 and some PIC12, PIC16 Families)
- Mid-range architectures (Some PIC12, PIC16 Families)
- Enhanced mid-range & high-performance architectures (PIC 18 Family)
- Advanced Architectures (PIC24 & PIC32 Families)

In addition, Microchip Technology Inc. also produces SAM Microcontrollers that are based on ARM Cortex M Processor architectures. PIC24 is a 16-bit microcontroller series with instruction code sizes of 24-Bit. PIC32 and SAM microcontroller that are based on ARM Cortex M are 32-bit Microcontrollers with 16-Bit or 32-Bit instruction code size.

PIC18 Microcontrollers are available in different packages like PDIP, PLCC, TQFP, SPDIP & SOIC. The number of pins vary based on package and range from 18 to 100 pins.

Interested readers may refer to Microchips documentation for additional details of various devices & families.

Table 3.2: Comparison of PIC18 Family

PIC18 Microcontroller Family

| Product    | Program Memory |         | Data Memory |              | I/O Ports | ADC 10-bit | MSSP                 | USART | Other    | CCP / Timers |          | Packages            | Pins  |
|------------|----------------|---------|-------------|--------------|-----------|------------|----------------------|-------|----------|--------------|----------|---------------------|-------|
|            | Type           | Bytes   | RAM Bytes   | EEPROM Bytes |           |            |                      |       |          | PWM          | 8/16-bit |                     |       |
| PIC18F1220 | FLASH          | 4K      | 256         | 256          | 16        | 7          | —                    | 1     | 6x PMM   | 1            | 1/3      | DIP SOIC, SSOP, QFN | 18    |
| PIC18F1320 | FLASH          | 8K      | 256         | 256          | 16        | 7          | —                    | 1     | 6x PMM   | 1            | 1/3      | DIP SOIC, SSOP, QFN | 18    |
| PIC18F2220 | FLASH          | 4K      | 512         | 256          | 23        | 10         | I <sup>2</sup> C/SPI | 1     | 6x PMM   | 2            | 1/3      | DIP SOIC, SSOP, QFN | 28    |
| PIC18F2320 | FLASH          | 8K      | 512         | 256          | 23        | 10         | I <sup>2</sup> C/SPI | 1     | 6x PMM   | 2            | 1/3      | DIP SOIC            | 28    |
| PIC18C242  | OTP            | 16K     | 512         | —            | 23        | 5          | I <sup>2</sup> C/SPI | 1     | —        | 2            | 1/3      | DIP SOIC            | 28    |
| PIC18C252  | OTP            | 32K     | 1536        | —            | 23        | 5          | I <sup>2</sup> C/SPI | 1     | —        | 2            | 1/3      | DIP SOIC            | 28    |
| PIC18F242  | FLASH          | 16K     | 512         | 256          | 23        | 5          | I <sup>2</sup> C/SPI | 1     | —        | 2            | 1/3      | DIP SOIC, SSOP      | 28    |
| PIC18F252  | FLASH          | 32K     | 1536        | 256          | 23        | 5          | I <sup>2</sup> C/SPI | 1     | —        | 2            | 1/3      | DIP SOIC, SSOP      | 28    |
| PIC18F258  | FLASH          | 32K     | 1536        | 256          | 22        | 5          | I <sup>2</sup> C/SPI | 1     | CAN 2.0B | 1            | 1/3      | DIP SOIC            | 28    |
| PIC18F4220 | FLASH          | 4K      | 512         | 256          | 34        | 13         | I <sup>2</sup> C/SPI | 1     | 6x PMM   | 2            | 1/3      | DIP TQFP QFN        | 40/44 |
| PIC18F4320 | FLASH          | 8K      | 512         | 256          | 34        | 13         | I <sup>2</sup> C/SPI | 1     | 6x PMM   | 2            | 1/3      | DIP TQFP QFN        | 40/44 |
| PIC18C442  | OTP            | 16K     | 512         | —            | 34        | 8          | I <sup>2</sup> C/SPI | 1     | —        | 2            | 1/3      | DIP PLCC, TQFP      | 40/44 |
| PIC18C452  | OTP            | 32K     | 1536        | —            | 34        | 8          | I <sup>2</sup> C/SPI | 1     | —        | 2            | 1/3      | DIP PLCC, TQFP      | 40/44 |
| PIC18F442  | FLASH          | 16K     | 512         | 256          | 34        | 8          | I <sup>2</sup> C/SPI | 1     | —        | 2            | 1/3      | DIP PLCC, TQFP      | 40/44 |
| PIC18F452  | FLASH          | 32K     | 1536        | 256          | 34        | 8          | I <sup>2</sup> C/SPI | 1     | —        | 2            | 1/3      | DIP PLCC, TQFP      | 40/44 |
| PIC18F458  | FLASH          | 32K     | 1536        | 256          | 33        | 5          | I <sup>2</sup> C/SPI | 1     | CAN 2.0B | 1            | 1/3      | DIP PLCC, TQFP      | 40/44 |
| PIC18C601  | —              | ROMless | 1536        | —            | 31        | 8          | I <sup>2</sup> C/SPI | 1     | —        | 2            | 1/3      | PLCC, TQFP          | 64/68 |
| PIC18C658  | OTP            | 32K     | 1536        | —            | 52        | 12         | I <sup>2</sup> C/SPI | 1     | CAN 2.0B | 2            | 1/3      | PLCC, TQFP          | 64/68 |
| PIC18F6520 | FLASH          | 32K     | 2048        | 1024         | 52        | 12         | I <sup>2</sup> C/SPI | 2     | —        | 5            | 2/3      | TQFP                | 64    |
| PIC18F6620 | FLASH          | 64K     | 3840        | 1024         | 52        | 12         | I <sup>2</sup> C/SPI | 2     | —        | 5            | 2/3      | TQFP                | 64    |
| PIC18F720  | FLASH          | 128K    | 3840        | 1024         | 52        | 12         | I <sup>2</sup> C/SPI | 2     | —        | 5            | 2/3      | TQFP                | 64    |
| PIC18C801  | —              | ROMless | 1536        | —            | 42        | 12         | I <sup>2</sup> C/SPI | 1     | —        | 2            | 1/3      | PLCC, TQFP          | 80/84 |
| PIC18C858  | OTP            | 32K     | 1536        | —            | 68        | 16         | I <sup>2</sup> C/SPI | 1     | CAN 2.0B | 2            | 1/3      | PLCC, TQFP          | 80/84 |
| PIC18F8520 | FLASH          | 32K     | 2048        | 1024         | 68        | 16         | I <sup>2</sup> C/SPI | 2     | EMA      | 5            | 2/3      | TQFP                | 80    |
| PIC18F8620 | FLASH          | 64K     | 3840        | 1024         | 68        | 16         | I <sup>2</sup> C/SPI | 2     | EMA      | 5            | 2/3      | TQFP                | 80    |
| PIC18F8720 | FLASH          | 128K    | 3840        | 1024         | 68        | 16         | I <sup>2</sup> C/SPI | 2     | EMA      | 5            | 2/3      | TQFP                | 80    |

Abbreviation: ADC = Analog-to-Digital Converter CCP = Capture/Compare/PWM I<sup>2</sup>C = Inter Integrated Circuit Bus PMM = Power Managed Mode  
 PWM = Pulse Width Modulation SPI = Serial Peripheral Interface USART = Universal Synchronous/Asynchronous Receiver/Transmitter

Source: Microchip Document: PIC18 Microcontroller Family - 30327b

**Table 3.3: Comparison of PIC18Fxx2 Series**

| Features                        | PIC18F242                                                                        | PIC18F252                                                                        | PIC18F442                                                                        | PIC18F452                                                                        |
|---------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Operating Frequency             | DC - 40 MHz                                                                      | DC - 40 MHz                                                                      | DC - 40 MHz                                                                      | DC - 40 MHz                                                                      |
| Program Memory (Bytes)          | 16K                                                                              | 32K                                                                              | 16K                                                                              | 32K                                                                              |
| Program Memory (Instructions)   | 8192                                                                             | 16384                                                                            | 8192                                                                             | 16384                                                                            |
| Data Memory (Bytes)             | 768                                                                              | 1536                                                                             | 768                                                                              | 1536                                                                             |
| Data EEPROM Memory (Bytes)      | 256                                                                              | 256                                                                              | 256                                                                              | 256                                                                              |
| Interrupt Sources               | 17                                                                               | 17                                                                               | 18                                                                               | 18                                                                               |
| I/O Ports                       | Ports A, B, C                                                                    | Ports A, B, C                                                                    | Ports A, B, C, D, E                                                              | Ports A, B, C, D, E                                                              |
| Timers                          | 4                                                                                | 4                                                                                | 4                                                                                | 4                                                                                |
| Capture/Compare/PWM Modules     | 2                                                                                | 2                                                                                | 2                                                                                | 2                                                                                |
| Serial Communications           | MSSP,<br>Addressable<br>USART                                                    | MSSP,<br>Addressable<br>USART                                                    | MSSP,<br>Addressable<br>USART                                                    | MSSP,<br>Addressable<br>USART                                                    |
| Parallel Communications         | —                                                                                | —                                                                                | PSP                                                                              | PSP                                                                              |
| 10-bit Analog-to-Digital Module | 5 input channels                                                                 | 5 input channels                                                                 | 8 input channels                                                                 | 8 input channels                                                                 |
| RESETS (and Delays)             | POR, BOR,<br>RESET Instruction,<br>Stack Full,<br>Stack Underflow<br>(PWRT, OST) | POR, BOR,<br>RESET Instruction,<br>Stack Full,<br>Stack Underflow<br>(PWRT, OST) | POR, BOR,<br>RESET Instruction,<br>Stack Full,<br>Stack Underflow<br>(PWRT, OST) | POR, BOR,<br>RESET Instruction,<br>Stack Full,<br>Stack Underflow<br>(PWRT, OST) |
| Programmable Low Voltage Detect | Yes                                                                              | Yes                                                                              | Yes                                                                              | Yes                                                                              |
| Programmable Brown-out Reset    | Yes                                                                              | Yes                                                                              | Yes                                                                              | Yes                                                                              |
| Instruction Set                 | 75 Instructions                                                                  | 75 Instructions                                                                  | 75 Instructions                                                                  | 75 Instructions                                                                  |
| Packages                        | 28-pin DIP<br>28-pin SOIC                                                        | 28-pin DIP<br>28-pin SOIC                                                        | 40-pin DIP<br>44-pin PLCC<br>44-pin TQFP                                         | 40-pin DIP<br>44-pin PLCC<br>44-pin TQFP                                         |

Source: Microchip Document: PIC18FXX2 Data Sheet – DS39564C

### Enhanced Mid-Range Architecture

PIC18 is enhanced mid-range family of microcontrollers. A comparison of PIC18 Family is presented in Table 3.2. The common PIC18 device series are PIC18Fxx2 and PIC18Fxx8. A comparison of PIC18Fxx2 series is presented in Table 3.3 and comparison of PIC18Fxx8 series is presented in Table 3.4.

**Table 3.4: Comparison of PIC18Fxx8 Series**

| Features                               |                               | PIC18F248                                                            | PIC18F258                                                            | PIC18F448                                                            | PIC18F458                                                            |
|----------------------------------------|-------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|
| Operating Frequency                    |                               | DC – 40 MHz                                                          | DC – 40 MHz                                                          | DC – 40 MHz                                                          | DC – 40 MHz                                                          |
| Internal Program Memory                | Bytes                         | 16K                                                                  | 32K                                                                  | 16K                                                                  | 32K                                                                  |
|                                        | # of Single-Word Instructions | 8192                                                                 | 16384                                                                | 8192                                                                 | 16384                                                                |
| Data Memory (Bytes)                    |                               | 768                                                                  | 1536                                                                 | 768                                                                  | 1536                                                                 |
| Data EEPROM Memory (Bytes)             |                               | 256                                                                  | 256                                                                  | 256                                                                  | 256                                                                  |
| Interrupt Sources                      |                               | 17                                                                   | 17                                                                   | 21                                                                   | 21                                                                   |
| I/O Ports                              |                               | Ports A, B, C                                                        | Ports A, B, C                                                        | Ports A, B, C, D, E                                                  | Ports A, B, C, D, E                                                  |
| Timers                                 |                               | 4                                                                    | 4                                                                    | 4                                                                    | 4                                                                    |
| Capture/Compare/PWM Modules            |                               | 1                                                                    | 1                                                                    | 1                                                                    | 1                                                                    |
| Enhanced Capture/Compare/PWM Modules   |                               | —                                                                    | —                                                                    | 1                                                                    | 1                                                                    |
| Serial Communications                  |                               | MSSP, CAN, Addressable USART                                         | MSSP, CAN, Addressable USART                                         | MSSP, CAN, Addressable USART                                         | MSSP, CAN, Addressable USART                                         |
| Parallel Communications (PSP)          |                               | No                                                                   | No                                                                   | Yes                                                                  | Yes                                                                  |
| 10-bit Analog-to-Digital Converter     |                               | 5 input channels                                                     | 5 input channels                                                     | 8 input channels                                                     | 8 input channels                                                     |
| Analog Comparators                     |                               | No                                                                   | No                                                                   | 2                                                                    | 2                                                                    |
| Analog Comparators VREF Output         |                               | N/A                                                                  | N/A                                                                  | Yes                                                                  | Yes                                                                  |
| Resets (and Delays)                    |                               | POR, BOR, RESET Instruction, Stack Full, Stack Underflow (PWRT, OST) | POR, BOR, RESET Instruction, Stack Full, Stack Underflow (PWRT, OST) | POR, BOR, RESET Instruction, Stack Full, Stack Underflow (PWRT, OST) | POR, BOR, RESET Instruction, Stack Full, Stack Underflow (PWRT, OST) |
| Programmable Low-Voltage Detect        |                               | Yes                                                                  | Yes                                                                  | Yes                                                                  | Yes                                                                  |
| Programmable Brown-out Reset           |                               | Yes                                                                  | Yes                                                                  | Yes                                                                  | Yes                                                                  |
| CAN Module                             |                               | Yes                                                                  | Yes                                                                  | Yes                                                                  | Yes                                                                  |
| In-Circuit Serial Programming™ (ICSP™) |                               | Yes                                                                  | Yes                                                                  | Yes                                                                  | Yes                                                                  |
| Instruction Set                        |                               | 75 Instructions                                                      | 75 Instructions                                                      | 75 Instructions                                                      | 75 Instructions                                                      |
| Packages                               |                               | 28-pin SPDIP<br>28-pin SOIC                                          | 28-pin SPDIP<br>28-pin SOIC                                          | 40-pin PDIP<br>44-pin PLCC<br>44-pin TQFP                            | 40-pin PDIP<br>44-pin PLCC<br>44-pin TQFP                            |

Source: Microchip Document: PIC18FXX8 Data Sheet - DS41159E

### 3.2 Comparison of 8051 & PIC18

One of the popular devices in PIC18 microcontrollers family is PIC18F458. The most popular and classical microcontroller is the Intel 8051. The comparison of Intel 8051 and Microchips PIC18F458 is given in Table 3.5

**Table 3.5: Comparison of 8051 & PIC18F458**

| Feature          |             | 8051/52                 | PIC 18F458                                                       |
|------------------|-------------|-------------------------|------------------------------------------------------------------|
| Architecture     |             | Harvard                 | Harvard                                                          |
| No of Bit Device |             | 8-Bit                   | 8-Bit                                                            |
| Program Memory   | Memory Size | 64K Max,<br>4K internal | 2M Max in PIC18,<br>Internal 32K Flash ROM<br>16384 Single words |
|                  | Address Bus | 16-Bit                  | 21-Bit                                                           |

|                                                              |             |                                                                                                              |                                                                                    |
|--------------------------------------------------------------|-------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|                                                              | Data Bus    | 8-Bit                                                                                                        | 16-Bit                                                                             |
| Data Memory                                                  | Memory Size | 64K Max (Ext),<br>128+SFR/256 Bytes internal                                                                 | 4K Max in PIC18,<br>GPR 1536 & SFR 256 Bytes<br>EEPROM 256 Bytes                   |
|                                                              | Address Bus | 8-Bit                                                                                                        | 12-Bit                                                                             |
|                                                              | Data Bus    | 8-Bit                                                                                                        | 8-Bit                                                                              |
| <b>Feature</b>                                               |             | <b>8051/52</b>                                                                                               | <b>PIC 18F458</b>                                                                  |
| Registers                                                    |             | Default Register A and SFR's Regs B, PC, SP, DPTR, PSW, 4 Banks of 8 Register: R0 to R7 & Peripheral related | Default Register WREG, PC, STKPTR, Status, File Registers & Peripheral related SFR |
| Timers                                                       |             | 2 or 3                                                                                                       | 4                                                                                  |
| I/O Pins                                                     |             | 32                                                                                                           | 34 Variations                                                                      |
| CCP/PWM                                                      |             | No                                                                                                           | Yes                                                                                |
| Serial Port USART                                            |             | Yes                                                                                                          | Yes                                                                                |
| Other Serial Protocols<br>Master Serial Port - SPI/I2C, CAN  |             | No                                                                                                           | Yes                                                                                |
| A/D Converter                                                |             | No                                                                                                           | Yes 10-Bit, 8-Channel                                                              |
| Analog Comparator                                            |             | No                                                                                                           | Yes                                                                                |
| Adv Features – Watchdog Timer, Brownout & Low voltage detect |             | No                                                                                                           | Yes                                                                                |
| Interrupts                                                   |             | 6                                                                                                            | 21                                                                                 |
| Instructions                                                 |             | 45                                                                                                           | 75                                                                                 |

### 3.3 Features of PIC 18F458

The noticeable features of PIC18F458 are

- Employs Harvard Architecture
- 8-Bit Microcontroller with High Performance RISC CPU
- Flash ROM Program memory of 32K or 16384 single instruction words with 21-Bit address bus and 16-Bit data bus & 16-Bit wide instruction.
- 31 Level dedicated stack
- Data memory of 1792 bytes (1536 bytes of GPR and 256 bytes of SFR) with 12-Bit address bus and 8-Bit data bus. In addition, it has 256 bytes of EEPROM Data memory.
- 75 Instructions with Flexible Destination File Register
- 8 x 8 Single-Cycle Hardware Multiplier

- Up to 10 MIPS operation
- DC – 40 MHz clock input
- 4 MHz-10 MHz oscillator/clock input with PLL active
- Peripherals like Timers, CCP, ECCP Module, ADC, Analog Comparator
- Serial Communication protocols USART, CAN, and Master Serial Port supporting SPI or I2C protocols.
- 5 Ports – Port A, B, C, D, E in the packages with 40 or more pins including a Parallel Slave Port.
- Advanced features like Watchdog Timer, Programmable Low Voltage & Brown out Detect and a Voltage Reference Module, Code Protection, In Circuit Serial Programming and debugging.
- 21 Interrupt Sources with 2 priority levels

### **3.4 Architecture of PIC18F458**

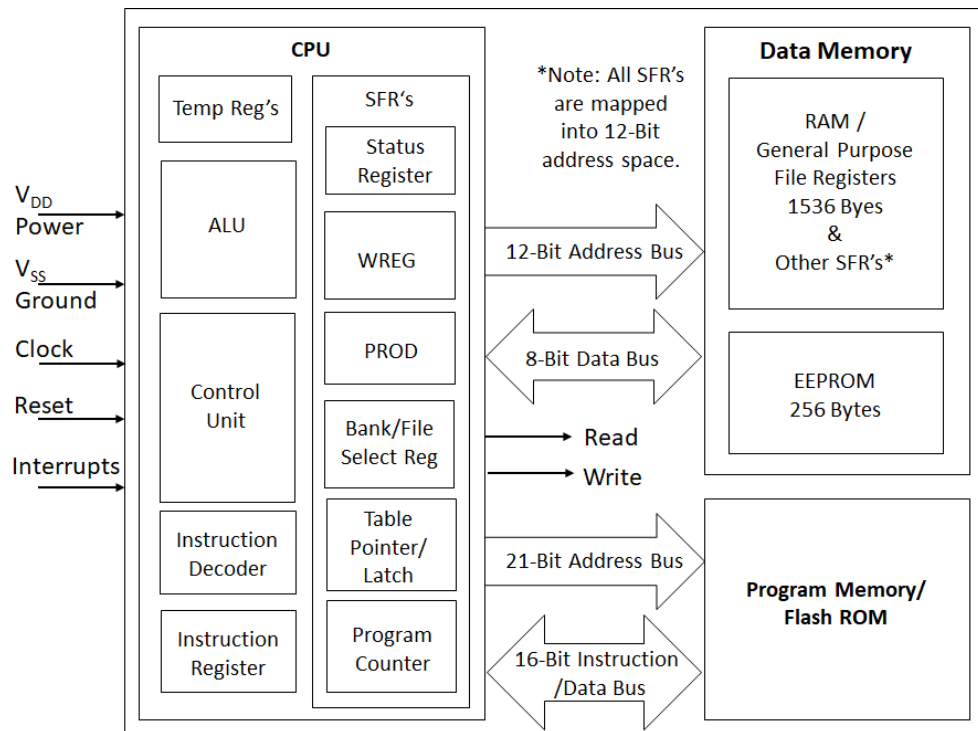
The PIC18F458 microcontroller employs Harvard architecture indicating that it has separate address and data bus each for program memory and data memory. The basic components of a microcontroller are presented in the block diagram of Processing Unit's interface with Memory that is shown in Fig 3.1.

The basic components are Central Processing Unit, ROM Program memory, RAM data memory with special function registers including WREG – Working Register, Status Register, Program Counter, File Select and Bank Select Registers. The program memory and data memory are connected to the processing unit through separate address bus and data buses and control bus (not shown in figure).

The central processing unit contains Arithmetic Logic Unit (ALU), Control Unit, Instruction Decoder, Instruction Register (IR).

Special function registers like Program Counter, WREG, Status Register are used by the CPU along with Bank Select and File Select registers. These registers are memory mapped register, i.e. They are allocated separate address space in RAM. In addition, various other special purpose registers are present that are used by CPU for configuration and operation of the different peripherals and memory in the microcontroller. The registers are discussed in their associated sections. The various components presented in Fig 3.1 are discussed here in the order of operation.





**Fig 3.1 Block Diagram of Processing Unit's interface with Memory**

### Program Counter

PIC microcontrollers have their program memory loaded with set of instructions to perform a sequence of tasks required for an application. An instruction counter is used to go through this sequence of tasks. It is called the Program Counter. The program counter is a 21-Bit register that holds the address of the next instruction to be executed. The PC register is readable and writable. The 21-Bit value is stored in combination of three 8-Bit registers PCU, PCH & PCL registers. PCU holds bits 16-21, PCH holds 8-15 & PCL holds 0-7 bits. The upper 3-bits in PCU are neglected. This can be observed in Fig 3.4.

The PCU and PCH are buffered through PCLATU and PCLATH respectively. The PCU and PCH are read and written through PCLATU & PCLATH when a PCL is read or written. A CALL, RCALL, GOTO and program branch instructions write to PC directly without PCLATH and PCLATU. The Least Significant Bit of PCL is fixed to a value of '0' for 16-Bit word alignment and the PC is incremented by 2 every time. The PCLATU, PCLATH & PCL are mapped in the address space of RAM.

### Instruction Register

Each instruction is coded with a group of bits that are used to configure and perform certain operations in the microcontroller. This group of bits is called an operation code and is loaded into instruction register from a specific location of program memory pointed by a

Program Counter. In simple terms the Instruction register holds the operation code of the instruction that needs to be executed. Here in PIC18 the instruction code is 16-Bit and so the Instruction register size is 16-Bit.

#### **Instruction Decoder**

Instruction decoder is a hardware component that is made up of combinational and sequential circuits. The purpose of the Instruction decoder is to decipher/decode the operations required for the respective instruction/operation code.

#### **Control Unit**

The control unit is hardware component that operates at scaled frequency of input crystal oscillator. It acquires the decoded information from instruction decoder and accordingly performs the respective operations. It provides the respective timing and control signals which act as activating or deactivating control signals to the different part of a microcontroller in a specific timing order.

#### **Arithmetic Logic Unit (ALU)**

The arithmetic and logic unit as the name suggests is used to perform arithmetic and logical operations. It has a combination of arithmetic and logical circuits that consists of adders, comparators, shifters etc. to perform the arithmetic operations such as addition, subtraction, multiplication and logical operations such as AND, OR, Exclusive OR etc. The ALU in PIC18 can perform 8-bit operations and is configured by the control unit to perform the respective operations based on the given instruction. The list of operations and their respective instructions are discussed along with instruction set.

#### **STATUS Register (Flag Register)**

The execution of arithmetic or logical operations generates some status signals. These are indicated through flags. The status register is the group of such flags that indicate the status of arithmetic and logical operations. The STATUS register is an 8-Bit register and is mapped in the address space of RAM. It contains the following flags: C – Carry, DC – Digital Carry, Z – Zero, OV - Overflow, N – Negative. The format of the status register is given in Fig 3.2. Appropriate examples of setting or resetting of the flags is demonstrated through the instructions in the programming section.

#### **Working Register(WREG)**

Working Register is the register that is used to perform any of the arithmetic and logical operations. It acts as source operand and may be used also a destination operand. WREG is generally called as an accumulator in Intel devices. PIC18 instruction set provide the flexibility to choose the destination register for storing the result of an operation. The WREG is an 8-Bit register and is mapped in the address space of RAM.

#### **Bank Select Register (BSR)**

The RAM in PIC18 is organized into different banks (16) for optimizing the instruction code. The Bank Select Register BSR is used to select a specific bank of RAM. It is

used for direct addressing of file registers in specific banks. The BSR is a 4-Bit register and is mapped in the address space of RAM, denoted as BSR<3:0>. It actually is an 8-Bit register with higher 4 bits unused and hence neglected. The banked organization of RAM is discussed in memory organization section.

### **File Select Register (FSR)**

The File Select Register is used for indirect access of RAM. These registers hold the register indirect address of RAM that is to be accessed. There are three File Select Registers – FSR0, FSR1 and FSR2. Each FSR is a 16-Bit register which is a combination of two 8-Bit registers High and Low i.e. FSR0H and FSR0L. Similarly, FSR1H & FSR1L for FSR1 and FSR2H & FSR2L for FSR2. The FSRs are mapped in the address space of RAM. The FSR's are used in auto increment or decrement modes through POSTINC, POSTDEC, INDF, PREINC & PLUSW commands that are also mapped in RAM address space. The usage of these is demonstrated through assembly language programming examples.

### **Buses**

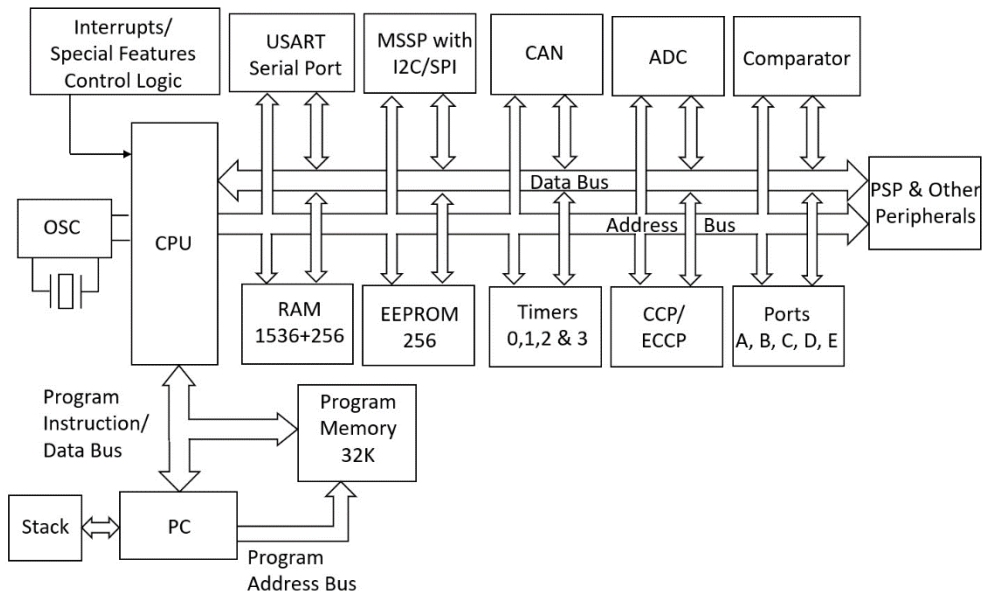
There are three types of buses in PIC18 microcontrollers. They are address, data and control buses. Since PIC18F458 employs a Harvard architecture, it has a set of the three buses for program memory and a set of the three buses for data memory.

**Address Bus:** PIC18 microcontrollers has separate unidirectional address bus from CPU to program and data memory, there is a separate address space for program memory and data memory. i.e. both the program memory and data memory can have the same address and yet stay unique from each other as they are separately accessed and controlled through separate buses. Address bus connected to Program Memory is a 21-Bit address bus and carries the address of a location in Flash ROM. So maximum addressing capability of Flash ROM is  $2^{21} = 2$  Mega Bytes. The address bus connected to Data Memory is a 12-Bit address bus and carries the address of EEPROM, RAMs GPR, SFR or Port I/O or peripheral. So maximum addressing capability of RAM is  $2^{12} = 4$  K Bytes.

**Data Bus:** Similar to Separate address buses but bi-directional to or from CPU, PIC18F458 has separate data bus for program and data memory. The data bus connected to Program Memory is a 16-Bit data bus and carries the instruction code or data constants from Flash ROM to the respective destination. The data bus connected to Data Memory is an 8-Bit data bus and carries the data to or from of EEPROM, RAMs GPR, SFR or Port I/O or peripheral.

**Control Bus:** The control bus constitutes the control signals generated by the central processing unit. These are used as activating/deactivating or read/write signals for the components in PIC18.

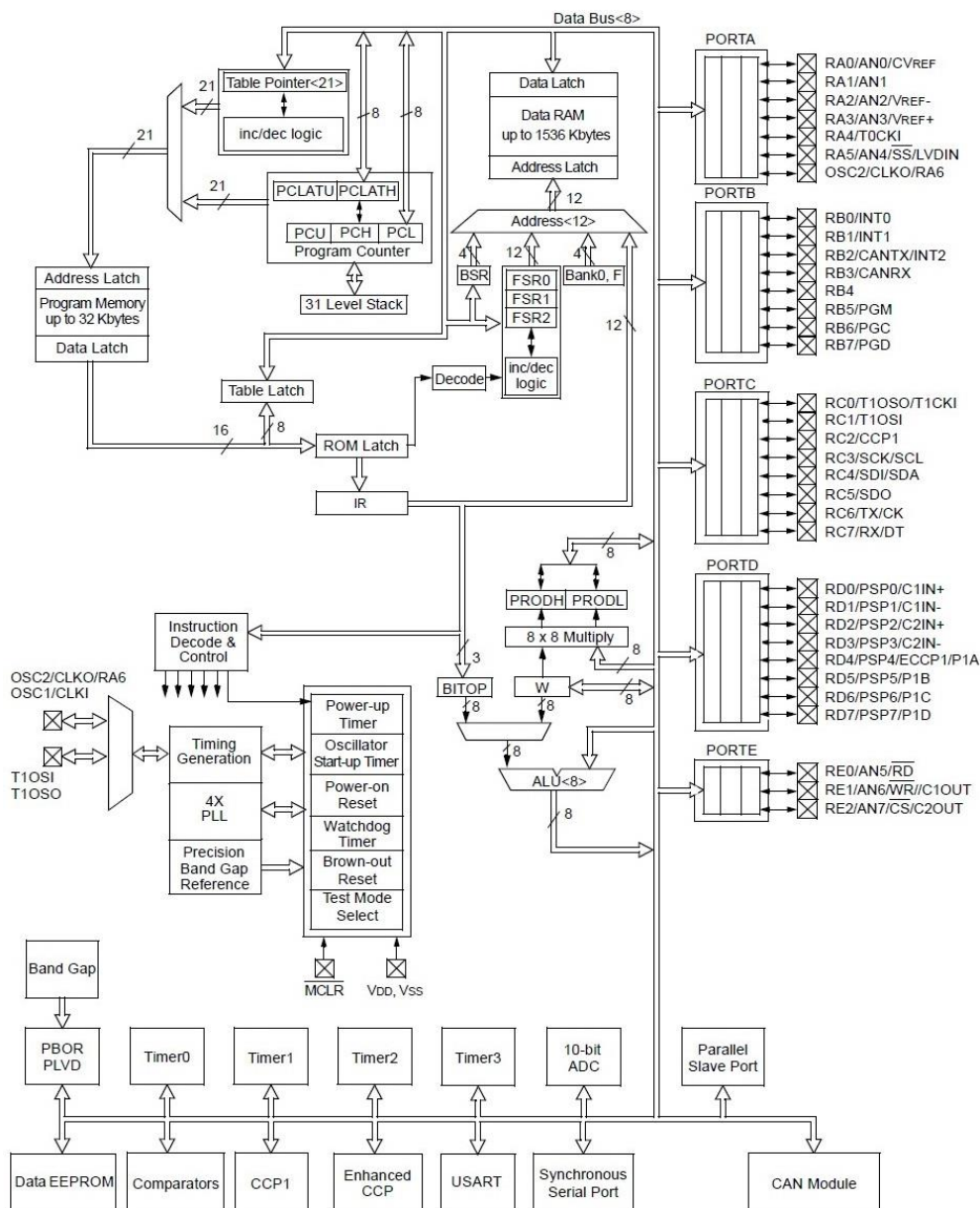
A simplified view of PIC18F458 Microcontroller that shows the interfacing of CPU with different peripherals and memory is presented in Fig 3.3. A detailed view of PIC18F458 Microcontroller that shows greater details and organization with different ports and special features is presented in Fig 3.4.



**Fig 3.3 Simplified View of PIC18F458 Microcontroller**

The PIC18F458 contains the following additional components.

- Dedicated 8x8-Bit Multiplier with 16-Bit result in registers PRODH (8-15 bit) and PRODL (0-7 bit) registers.
- 31 Level dedicated stack to store program addresses temporarily.
- Table Latch & a 21-Bit Table Pointer to access table data in program memory.
- 1 ADC with 10-Bit resolution and 8-Channel Input
- 2 Analog Comparators
- Voltage Reference Module
- 1 Master Serial Port configurable that can support Serial Peripheral Interface (SPI) or Inter Integrated Circuit (I2C) protocols
- Controller Area Network - CAN Serial communication protocol
- 1 Parallel Slave Port.
- Special features like Watchdog Timer, Programmable Low Voltage & Brown out Detect.
- It also has 5 Ports in the packages that have over 40 pins. i.e. Port A with 7 Pins, Port B, C, D with 8 Pins and Port E with 3 Pins.
- 21 Interrupt Sources with 2 priority levels.
- 4 Timers (i.e. One 8-Bit Timer 2 and Three 16-Bit Timers: Timer 0, 1, and 3).
- 1 Capture Compare Module
- 1 Enhanced Capture Compare Module
- 1 USART



**Fig 3.4 Detailed Diagram of PIC18F458**

Source: Microchip Document: PIC18FXX8 Data Sheet - DS41159E

The overview of the peripherals is presented in Peripherals sections and details of the same are discussed in the corresponding peripheral sections.

### STATUS Register Format

|     |     |     |       |       |       |       |       |
|-----|-----|-----|-------|-------|-------|-------|-------|
| U-0 | U-0 | U-0 | R/W-x | R/W-x | R/W-x | R/W-x | R/W-x |
| —   | —   | —   | N     | OV    | Z     | DC    | C     |

bit 7

bit 0

| Bit No  | Bit           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bit 7-5 | Unimplemented | Read as '0'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| bit 4   | N             | <b>Negative bit</b><br>This bit is used for signed arithmetic (2's complement). It indicates whether the result of the ALU operation was negative (ALU MSB = 1).<br>1 = Result was negative<br>0 = Result was positive                                                                                                                                                                                                                                                                         |
| bit 3   | OV            | <b>Overflow bit</b><br>This bit is used for signed arithmetic (2's complement). It indicates an overflow of the 7-bit magnitude which causes the sign bit (bit 7) to change state.<br>1 = Overflow occurred for signed arithmetic (in this arithmetic operation)<br>0 = No overflow occurred                                                                                                                                                                                                   |
| bit 2   | Z             | <b>Zero bit</b><br>1 = The result of an arithmetic or logic operation is zero<br>0 = The result of an arithmetic or logic operation is not zero                                                                                                                                                                                                                                                                                                                                                |
| bit 1   | DC            | <b>Digit Carry/Borrow bit</b><br>For ADDWF, ADDLW, SUBLW and SUBWF instructions:<br>1 = A carry-out from the 4th low-order bit of the result occurred<br>0 = No carry-out from the 4th low-order bit of the result<br><b>Note:</b> For <u>Borrow</u> , the polarity is reversed.<br>A subtraction is executed by adding the 2's complement of the second operand. For rotate (RRCF, RRNCF, RLCF and RLNCF) instructions, this bit is loaded with either bit 4 or bit 3 of the source register. |

|       |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bit 0 | C | <b>Carry/Borrow bit</b><br>For ADDWF, ADDLW, SUBLW and SUBWF instructions:<br>1 = A carry-out from the Most Significant bit of the result occurred<br>0 = No carry-out from the Most Significant bit of the result occurred<br><b>Note:</b> For <u>Borrow</u> , the polarity is reversed.<br>A subtraction is executed by adding the 2's complement of the second operand. For rotate (RRF, RLF) instructions, this bit is loaded with either the high or low-order bit of the source register. |
|-------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Fig 3.2 Format of STATUS Register**

### 3.5 Memory organization

The PIC18 microcontrollers makes use of Flash program memory, EEPROM data memory and RAM data memory that has GPR's - General Purpose Registers and SFR's - Special Function Registers. It additionally has separate storage for stack.

As already discussed, earlier PIC18 employs a Harvard architecture with separate buses for the data and program memories. This allows for parallel transfer of instruction code from Flash Program ROM and data from RAM at the same time which results in faster execution of application programs.

#### 3.5.1 Program Memory - ROM

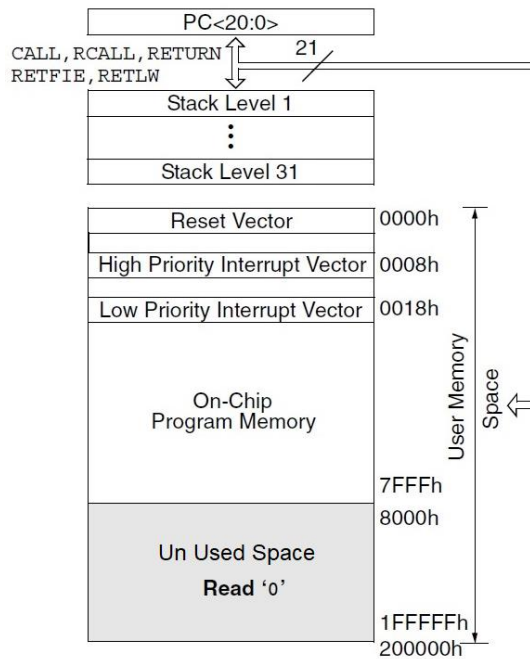
ROM is Read Only Memory. The purpose of ROM is to store the instructions of an application program. In PIC18 it is interfaced with CPU through 21-Bit address bus and 16-Bit data bus. The PIC18C microcontrollers have EEPROM program memory and PIC18F microcontrollers have Flash Program memory. The Flash ROM's address range of PIC18's is 000000 – 1FFFFFFH with total of 2M bytes and the range of PIC18F458 is 000000 - 007FFFH with total of 32K bytes or 16K single word instructions. The unused addresses from 008000 to 1FFFFFFH are read as 0. Of the 75 instructions of PIC18F458, all the instructions are 16-Bit wide except for the 4 instructions which are 32-Bit wide i.e. MOVFF, LFSR, CALL and GOTO instructions. The vector addresses for Reset are 000000H and 000008H for High priority and 000018H for Low priority interrupts. Program Counter gets incremented 2 times for 16-Bit word instructions and 4 times for a 32-Bit word instruction. The steps required to program the Flash program memory are discussed in detail in EEPROM, Flash Memory chapter.

#### 3.5.2 Stack Memory

Two types of stacks are present in PIC18F458. 1) Program memory stack with 31 levels and 2) Fast Stack with one level to store WREG, Status and BSR registers. PIC18F458

does not employ stack for data memory. The one level fast stack is only for WREG, STATUS & BSR. General purpose memory needs to be used for temporary storage of data.

#### Program memory stack with 31 levels



**Fig 3.5 Organization of Flash ROM in PIC18F458**

Source: Microchip Document: PIC18FXX8 Data Sheet - DS41159E

Traditional stack memory is used to temporarily store the address of program memory during application program execution. The PIC18F458 has a dedicated 31 level stack which is not part of RAM or ROM. Each location/level in stack has the capability to store a 21-bit word sized Flash ROM address value. The stack memory is used to store the PC content when interrupt is acknowledged or when instructions such as PUSH, POP, CALL, RCALL, RETURN & RETLW are executed. The stack pointer STKPTR register provides the status and usage of the stack. The STKPTR register is also mapped in the address space of RAM. The format of the STKPTR register is presented in Fig 3.6.

The bit STKFUL in the STKPTR register indicates whether stack is full. It is set to 1 when more than 31 PUSH operations occur else it is 0. This bit needs to be cleared by user software or by a POR. STKFUNF indicates whether stack is empty. It is set to 1 when more than 31 POP operations occur. This bit needs to be cleared by user software or by a POR.

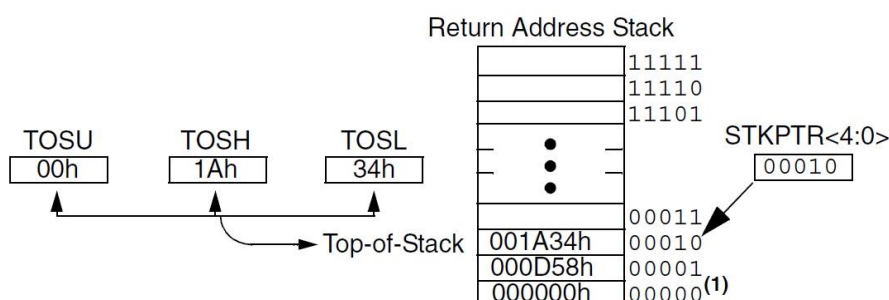
#### STKPTR – Stack Pointer Register Format

| R/C-0  | R/C-0   | U-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 | R/W-0 |
|--------|---------|-----|-------|-------|-------|-------|-------|
| STKFUL | STKFUNF | —   | SP4   | SP3   | SP2   | SP1   | SP0   |
| bit 7  |         |     |       |       |       |       | bit 0 |



| Bit No  | Bit           | Description                                                                                                                            |
|---------|---------------|----------------------------------------------------------------------------------------------------------------------------------------|
| bit 7   | STKFUL        | <b>Stack Full Flag bit</b><br>1 = Stack became full or overflowed<br>0 = Stack has not become full or overflowed                       |
| bit 6   | STKUNF        | <b>Stack Underflow Flag bit</b><br>1 = Stack underflow occurred<br>0 = Stack underflow did not occur                                   |
| bit 5   | Unimplemented | Read as '0'                                                                                                                            |
| bit 4-0 | SP4:SP0       | <b>Stack Pointer Location bits</b><br><b>Note:</b> Bit 7 and bit 6 need to be cleared following a stack underflow or a stack overflow. |

**Fig 3.6 Stack Pointer STKPTR Register Format**



**Note 1:** No RAM associated with this address; always maintained '0's.

**Fig 3.7 Return Address Stack and associated Registers**

Source: Microchip Document: PIC18FXX8 Data Sheet - DS41159E

The 31 level stack registers are identified using 5-bits SP4-SP0. They indicate the register/level number of stack i.e. the stack address where PC contents are stored. These bits are initialized to 00000b after all Resets. The value that is stored in the level pointed by SP4-SP0 is called the top of the stack. Three registers TOSU, TOSH and TOSL are used to access the contents of the stack location indicated by the SP4-SP0 bits of the STKPTR register.

A CALL type instruction creates a push operation onto the stack. It causes the level number in Stack Pointer to first increment and then the contents of the PC are stored into the top of the Stack location of RAM and stack. A RETURN type instruction creates a pop operation from the stack. The register/level number of STKPTR points to stack location and the contents of it are transferred to PC. The level number in Stack Pointer is then decremented. The top of the stack would now contain the value of new pointer. A demonstration of stack during Return is presented in Fig 3.7.

The Stack Pointer and data in the top of the stack is readable and writable and allows for implementing a software stack. A configuration bit STVREN is a programmable bit. Setting this bit causes a device reset when an overflow or underflow of stack occurs.

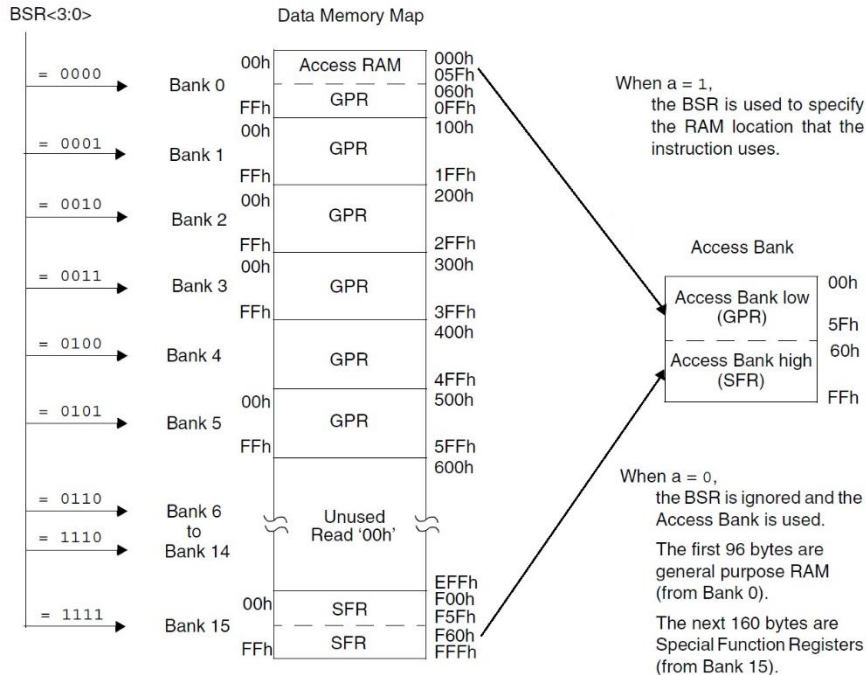
### Fast Register Stack with one level

The Fast register stack provides fast return for interrupts and call operations. This feature allows to store WREG, Status and BSR registers in one level fast stack. The registers are automatically stored when an interrupt occurs or when a fast access call is made. When a fast return is made the stack contents are copied/restored into the respective registers if a fast stack call is used earlier to store the respective registers. This one level stack is not readable or writeable. When nesting of interrupts occurs ( i.e while in low priority interrupt ISR, if a high priority interrupt occurs) the fast register stack is overwritten. The global interrupts should be disabled to prevent inadvertent stack operations.

### 3.5.3 Data Memory (General Purpose & Special Function Registers)

RAM is a read and write memory and the purpose of RAM is to store data. Hence it is called as data memory. It is interfaced with CPU through 12-Bit address bus and 8-Bit data bus. The data memory's address range of PIC18 series in general is 000 – FFFH with total of 4K bytes.

The range of PIC18F458 is 000 - 5FFH with total of 1536 bytes and 256 bytes of SFRs from F00H to FFFH. The unused addresses from 600 to EFFH are read as 0. The memory map of Data memory is presented in Fig 3.7. Each memory location in RAM is called as a file register and is denoted as 'f'.



**Fig 3.7 Organization of RAM in PIC18F458**

Source: Microchip Document: PIC18FXX8 Data Sheet - DS41159E

The data memory is divided into 16 banks, each with 256bytes. Additionally, there is a default bank named access bank. Access bank is a combination of GPR with locations ranging from 000 to 05FH and SFRs ranging from F60 to FFFH. In a PIC18F452 the access bank is combination of GPR with locations ranging from 000 to 07FH and SFRs ranging from F80 to FFFH. The Bank Select Register BSR<3:0> is used to select the corresponding bank. The MOVLB instruction is used to load the BSR register with the 4-Bit bank number. The instructions of PIC18 are provided with the option to select between the default access bank or one of the 16 banks. The option denoted as 'a' selects access bank when the value is 0 and selects the banks (0-F) based on BSR register when the 'a' value is 1. MOVFF instruction ignores the BSR bits.

The banking concept is a design technique employed by PIC microcontrollers to reduce the number of address bits that need to be specified along with the instructions. For a memory with 12-Bit address, traditionally, the number of bits required for addressing in instructions is 12-Bits, but using banking concept, in the case of PIC18, with 16 banks the addressable bits required to be used in instruction is brought down to 8-bits.

When accessing program or data it is expected that if one location is accessed, the following access most likely occurs in the vicinity of the earlier accessed location. i.e accesses are made in a specific locality. Hence if the locality is fixed the following accesses generally does not require full address. The following example demonstrates this concept.

Consider 8 buildings, each with 4 floors and 8 rooms per floor. So, a total of 256 rooms is available. If a reference needs to be made to a room, it can be referred by a unique number where room number can range from 1 to 256 requiring 8-bit address. If a person is already in a specific building(locality) and referring to a room in same building, there are only 32 rooms in that building which requires only 5-bits. So now the number of addressable bits required is reduced by 3-bits. Here the number of buildings associate to number of banks, and each floor may refer to group of 16 locations, like  $0x - Fx$  where x may be associated with a room or one of 16 locations.

So once a bank is selected the following instructions if accessing within the same bank does not require the full address bits and only require the bits that are required to access with in that bank.

Such an organization helps in reduction of number of bits required for addressing a location, number of bits required for an instruction which ultimately reduces the program code size. When addressed using full 12- bit address value in instructions (neglecting operation code), 100 instructions require 1200 bits.

| Address | Name     | Address | Name           | Address | Name            | Address | Name         |
|---------|----------|---------|----------------|---------|-----------------|---------|--------------|
| F00h    | RXF0SIDH | F20h    | TXB2CON        | F40h    | TXB0CON         | F60h    | RXB0CON (4)  |
| F01h    | RXF0SIDL | F21h    | TXB2SIDH       | F41h    | TXB0SIDH        | F61h    | RXB0SIDH (4) |
| F02h    | RXF0EIDH | F22h    | TXB2SIDL       | F42h    | TXB0SIDL        | F62h    | RXB0SIDL (4) |
| F03h    | RXF0EIDL | F23h    | TXB2EIDH       | F43h    | TXB0EIDH        | F63h    | RXB0EIDH (4) |
| F04h    | RXF1SIDH | F24h    | TXB2EIDL       | F44h    | TXB0EIDL        | F64h    | RXB0EIDL (4) |
| F05h    | RXF1SIDL | F25h    | TXB2DLC        | F45h    | TXB0DLC         | F65h    | RXB0DLC (4)  |
| F06h    | RXF1EIDH | F26h    | TXB2D0         | F46h    | TXB0D0          | F66h    | RXB0D0 (4)   |
| F07h    | RXF1EIDL | F27h    | TXB2D1         | F47h    | TXB0D1          | F67h    | RXB0D1 (4)   |
| F08h    | RXF2SIDH | F28h    | TXB2D2         | F48h    | TXB0D2          | F68h    | RXB0D2 (4)   |
| F09h    | RXF2SIDL | F29h    | TXB2D3         | F49h    | TXB0D3          | F69h    | RXB0D3 (4)   |
| F0Ah    | RXF2EIDH | F2Ah    | TXB2D4         | F4Ah    | TXB0D4          | F6Ah    | RXB0D4 (4)   |
| F0Bh    | RXF2EIDL | F2Bh    | TXB2D5         | F4Bh    | TXB0D5          | F6Bh    | RXB0D5 (4)   |
| F0Ch    | RXF3SIDH | F2Ch    | TXB2D6         | F4Ch    | TXB0D6          | F6Ch    | RXB0D6 (4)   |
| F0Dh    | RXF3SIDL | F2Dh    | TXB2D7         | F4Dh    | TXB0D7          | F6Dh    | RXB0D7 (4)   |
| F0Eh    | RXF3EIDH | F2Eh    | CANSTATRO4 (5) | F4Eh    | CANSTATRO 2 (5) | F6Eh    | CANSTAT      |
| F0Fh    | RXF3EIDL | F2Fh    | —              | F4Fh    | —               | F6Fh    | CANCON       |
| F10h    | RXF4SIDH | F30h    | TXB1CON        | F50h    | RXB1CON         | F70h    | BRGCON1      |
| F11h    | RXF4SIDL | F31h    | TXB1SIDH       | F51h    | RXB1SIDH        | F71h    | BRGCON2      |
| F12h    | RXF4EIDH | F32h    | TXB1SIDL       | F52h    | RXB1SIDL        | F72h    | BRGCON3      |
| F13h    | RXF4EIDL | F33h    | TXB1EIDH       | F53h    | RXB1EIDH        | F73h    | CIOCON       |
| F14h    | RXF5SIDH | F34h    | TXB1EIDL       | F54h    | RXB1EIDL        | F74h    | COMSTAT      |
| F15h    | RXF5SIDL | F35h    | TXB1DLC        | F55h    | RXB1DLC         | F75h    | RXERRCNT     |
| F16h    | RXF5EIDH | F36h    | TXB1D0         | F56h    | RXB1D0          | F76h    | TXERRCNT     |
| F17h    | RXF5EIDL | F37h    | TXB1D1         | F57h    | RXB1D1          | F77h    | —            |
| F18h    | RXM0SIDH | F38h    | TXB1D2         | F58h    | RXB1D2          | F78h    | —            |
| F19h    | RXM0SIDL | F39h    | TXB1D3         | F59h    | RXB1D3          | F79h    | —            |
| F1Ah    | RXM0EIDH | F3Ah    | TXB1D4         | F5Ah    | RXB1D4          | F7Ah    | —            |
| F1Bh    | RXM0EIDL | F3Bh    | TXB1D5         | F5Bh    | RXB1D5          | F7Bh    | —            |
| F1Ch    | RXM1SIDH | F3Ch    | TXB1D6         | F5Ch    | RXB1D6          | F7Ch    | —            |
| F1Dh    | RXM1SIDL | F3Dh    | TXB1D7         | F5Dh    | RXB1D7          | F7Dh    | —            |
| F1Eh    | RXM1EIDH | F3Eh    | CANSTATRO3 (5) | F5Eh    | CANSTATRO 1 (5) | F7Eh    | —            |
| F1Fh    | RXM1EIDL | F3Fh    | —              | F5Fh    | —               | F7Fh    | —            |

| Address | Name         | Address | Name            | Address | Name        | Address | Name         |
|---------|--------------|---------|-----------------|---------|-------------|---------|--------------|
| F80H    | PORTA        | FA0H    | PIE2            | FC0H    | —           | FE0H    | BSR          |
| F81H    | PORTB        | FA1H    | PIR2            | FC1H    | ADCON1      | FE1H    | FSR1L        |
| F82H    | PORTC        | FA2H    | IPR2            | FC2H    | ADCON0      | FE2H    | FSR1H        |
| F83H    | PORTD<br>(3) | FA3H    | PIE3            | FC3H    | ADRESL      | FE3H    | PLUSW1(2)    |
| F84H    | PORTE<br>(3) | FA4H    | PIR3            | FC4H    | ADRESH      | FE4H    | PREINC1(2)   |
| F85H    | —            | FA5H    | IPR3            | FC5H    | SSPCON2     | FE5H    | POSTDEC1(2)  |
| F86H    | —            | FA6H    | EECON1          | FC6H    | SSPCON1     | FE6H    | POSTINC1(2)  |
| F87H    | —            | FA7H    | EECON2          | FC7H    | SSPSTAT     | FE7H    | INDF1<br>(2) |
| F88H    | —            | FA8H    | EEDATA          | FC8H    | SSPADD      | FE8H    | WREG         |
| F89H    | LATA         | FA9H    | EEADR           | FC9H    | SSPBUF      | FE9H    | FSR0L        |
| F8AH    | LATB         | FAAH    | —               | FCAH    | T2CON       | FEAH    | FSR0H        |
| F8BH    | LATC         | FABH    | RCSTA           | FCBH    | PR2         | FEBH    | PLUSW0(2)    |
| F8CH    | LATD (3)     | FACH    | TXSTA           | FCCH    | TMR2        | FECH    | PREINC0(2)   |
| F8DH    | LATE (3)     | FADH    | TXREG           | FCDH    | T1CON       | FEDH    | POSTDEC0(2)  |
| F8EH    | —            | FAEH    | RCREG           | FCEH    | TMR1L       | FEEH    | POSTINC0(2)  |
| F8FH    | —            | FAFH    | SPBRG           | FCFH    | TMR1H       | FEFH    | INDF0(2)     |
| F90H    | —            | FB0H    | —               | FD0H    | RCON        | FF0H    | INTCON3      |
| F91H    | —            | FB1H    | T3CON           | FD1H    | WDTCON      | FF1H    | INTCON2      |
| F92H    | TRISA        | FB2H    | TMR3L           | FD2H    | LVDCON      | FF2H    | INTCON       |
| F93H    | TRISB        | FB3H    | TMR3H           | FD3H    | OSCCON      | FF3H    | PRODL        |
| F94H    | TRISC        | FB4H    | CMCON (3)       | FD4H    | —           | FF4H    | PRODH        |
| F95H    | TRISD(3)     | FB5H    | CVRCON (3)      | FD5H    | T0CON       | FF5H    | TABLAT       |
| F96H    | TRISE(3)     | FB6H    | ECCPAS (3)      | FD6H    | TMR0L       | FF6H    | TBLPTRL      |
| F97H    | —            | FB7H    | ECCP1DEL<br>(3) | FD7H    | TMR0H       | FF7H    | TBLPTRH      |
| F98H    | —            | FB8H    | —               | FD8H    | STATUS      | FF8H    | TBLPTRU      |
| F99H    | —            | FB9H    | —               | FD9H    | FSR2L       | FF9H    | PCL          |
| F9AH    | —            | FBAH    | ECCP1CON<br>(3) | FDAH    | FSR2H       | FFAH    | PCLATH       |
| F9BH    | —            | FBBH    | ECCPR1L (3)     | FDBH    | PLUSW2(2)   | FFBH    | PCLATU       |
| F9CH    | —            | FBCH    | ECCPR1H (3)     | FDCH    | PREINC2(2)  | FFCH    | STKPTR       |
| F9DH    | PIE1         | FBDH    | CCP1CON         | FDDH    | POSTDEC2(2) | FFDH    | TOSL         |
| F9EH    | PIR1         | FBEH    | CCPR1L          | FDEH    | POSTINC2(2) | FFEH    | TOSH         |
| F9FH    | IPR1         | FBFH    | CCPR1H          | FDFH    | INDF2(2)    | FFFH    | TOSU         |

**Table 3.6 Special Function Register Map**

- Note
- 1: The symbol — indicates Unimplemented registers and are read as '0'.
  - 2: This is not a physical register.
  - 3: These registers are not implemented on the PIC18F248 and PIC18F258.

4: Contents of register are dependent on WIN2:WIN0 bits in CANCON register.

5: CANSTAT register is repeated in these locations to simplify application firmware.

Unique names are given for each instance of the CANSTAT register due to the Microchip header file requirement.

The same when accessed within a bank/locality requires bank selection bits of 4 (BSR register value to select the respective bank) followed by 8-bit value in each of the 100 instructions totalling to 804 bits. This banking concept saves around 50 bytes for 100 instructions. A similar concept is implemented as segments in 8086 and 4 register banks in popular 8051 microcontroller and various other devices.

Each file register has 12-bit location address in the data memory and can be accessed in 3 different ways. 1) Direct addressing bank wise using 4-bit BSR based locality with 8-bit file address 2) Direct addressing using full 12-bit address (MOVFF instruction) 3) Indirect addressing using FSR. The FSR registers can be addressed using the names of registers or the address of SFR through access bank.

PIC18F458 is similar to 8051 in different aspects. Every register except PC in 8051 is address mapped. i.e. Register can be accessed using a register name or by an address and accessed directly or indirectly. PIC18F458 does not have general purpose registers as such like 8051 which has R0-R7 etc. but general-purpose memory is present. This general-purpose memory is address mapped and uses bank based 8-bit address or non-banked 12-bit address. Hence PIC18F458 devices does not name these by a specific register name for general purpose registers. They are just called as File registers and accessed using their address directly or indirectly. Special function registers though are named and are also address mapped similar to 8051.

Different ways of accessing the file registers is discussed in addressing modes and instructions section. The different SFR's - Special Function Registers and their address mapping in RAM is presented in table 3.6. The most used SFRs are classified based on their functionality, usage or an associated peripheral and are presented in table 3.7. The details of the SFRs are discussed in the associated sections.

### **3.5.4 Data EEPROM**

EEPROM is Electrically Erasable Programmable Read Only Memory that is used for frequent accesses. EEPROM is a separate memory and is not part of data memory space. It is different from Flash ROM as each location is accessible for read or write operations byte wise whereas a flash ROM is only accessed block wise. The PIC18f458 has 256 bytes of EEPROM with address range from 00H to FFH.

EEPROM is accessed or controlled through indirect addressing using 4 SFR's associated with EEPROM. i.e. EECON1, EECON2, EEDATA and EEADR. The EEADR register holds the address of location in EEPROM memory and EEDATA holds the data for the respective location. EECON1 and EECON2 are used for control purposes. The complete

details on the respective register formats and usage of EEPROM is presented later in EEPROM section.

| Main   | Bank/<br>Indirect | Indirect<br>Reg; Not a<br>Physical<br>Reg | Port  | Table   | Interrupt | Timer | ADC    | Clock<br>WDT LV | CCP      | EEPROM | USART | Master<br>Serial | Comparator |
|--------|-------------------|-------------------------------------------|-------|---------|-----------|-------|--------|-----------------|----------|--------|-------|------------------|------------|
| PCLATH | BSR               | INDF0                                     | TRISA | TABLAT  | INTCON    | T0CON | ADCON0 | OSCCON          | CCP1CON  | EEADR  | TXSTA | SSPADD           | CMCON      |
| PCLATU | FSR0H             | INDF1                                     | TRISB | TBLPTRH | INTCON2   | T1CON | ADCON1 | LVDCON          | CCPR1H   | EECON1 | RCSTA | SSPBUF           | CVRCON     |
| PCL    | FSR0L             | INDF2                                     | TRISC | TBLPTRL | INTCON3   | T2CON | ADRESH | WDTCON          | CCPR1L   | EECON2 | TXREG | SSPCON1          |            |
| WREG   | FSR1H             | POSTDEC0                                  | TRISD | TBLPTRU | IPR1      | T3CON | ADRESL |                 | ECCP1CON | EEDATA | RCREG | SSPCON2          |            |
| STATUS | FSR1L             | POSTDEC1                                  | TRISE |         | IPR2      | TMR0H |        |                 | ECCPIDEL |        | SPBRG | SSPSTAT          |            |
| PRODH  | FSR2H             | POSTDEC2                                  | LATA  |         | IPR3      | TMR0L |        |                 | ECCPAS   |        |       |                  |            |
| PRODL  | FSR2L             | POSTINC0                                  | LATB  |         | PIE1      | TMR1H |        |                 | ECCPR1H  |        |       |                  |            |
| TOSH   |                   | POSTINC1                                  | LATC  |         | PIE2      | TMR1L |        |                 | ECCPR1L  |        |       |                  |            |
| TOSL   |                   | POSTINC2                                  | LATD  |         | PIE3      | TMR2  |        |                 |          |        |       |                  |            |
| TOSU   |                   | PREINC0                                   | LATE  |         | PIR1      | TMR3H |        |                 |          |        |       |                  |            |
| STKPTR |                   | PREINC1                                   | PORTA |         | PIR2      | TMR3L |        |                 |          |        |       |                  |            |
|        |                   | PREINC2                                   | PORTB |         | PIR3      | PR2   |        |                 |          |        |       |                  |            |
|        |                   | PLUSW0                                    | PORTC |         | RCON      |       |        |                 |          |        |       |                  |            |
|        |                   | PLUSW1                                    | PORTD |         |           |       |        |                 |          |        |       |                  |            |
|        |                   | PLUSW2                                    | PORTE |         |           |       |        |                 |          |        |       |                  |            |

**Table 3.7 Summary of Registers - Functionality Wise (Excluding CAN, Configuration Registers etc)**

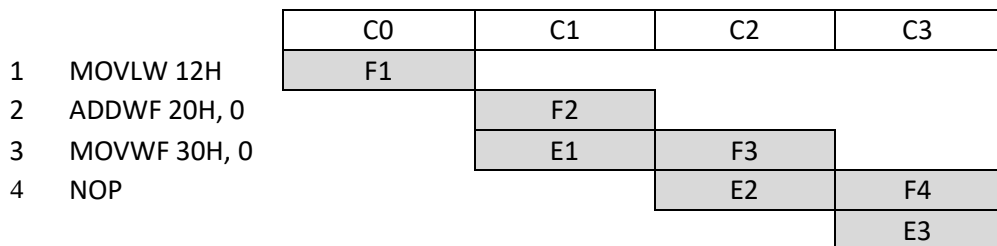
### 3.6 Pipelining & Instruction cycle

A microcontroller performs its operations based on timing cycle derived through the clock from crystal oscillator as a sequence of steps that are completed in different hardware units. To increase the efficiency hardware is organized as sequential units so that units process its tasks and work independent of one another or work in parallel.

#### Pipelining

The process of parallelising or overlapping of different hardware units to perform the required tasks is known as pipelining. The three common pipeline units are Fetch, Decode and execution units. Fetch unit perform the process of getting the instruction code from program memory to instruction register. Decoding unit perform the process of determining the required operations for the instruction. Execution unit performs the execution of the required operations.

The PIC18F458 employs a 2-Stage pipeline where stage 1 involves Fetch unit while decode and execution unit form the stage 2. Fetching of instruction 1 is carried out by stage 1 unit during the first clock cycle C0 and during second clock cycle C1, stage 1 unit performs fetching of second instruction in parallel with stage 2 unit that performs decode and execute of first instruction. The pipeline instruction flow for 3-instruction sequence is shown in Fig 3.8.



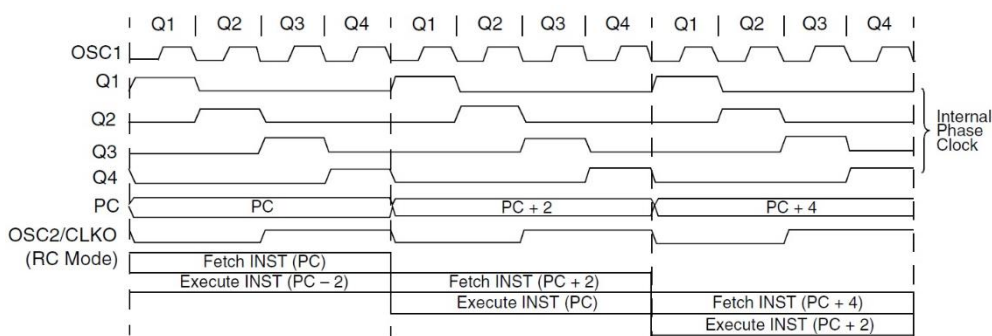
**Fig 3.8 Pipeline instruction flow for 3-instruction sequence in PIC18F458**

#### Instruction Cycle

Cycle time is the smallest time required to perform a sub operation. Machine cycle time is the time required or the number of clock cycles required to perform an operation. An instruction is a combination of one or more operations. The instruction cycle time is the number of machine cycles required for execute an instruction.

The PIC18F458 uses the oscillator signal with an internal divider and generates 4 Q cycle clocks denoted as Q1, Q2, Q3 and Q4 that are non-overlapping quadrature clock signals. In PIC terminology instruction cycle consists of the 4 cycles Q1 through Q4. They are shown in Fig 3.9. Most of the PIC18 instructions are 16-Bit or 1-word size and are accessed in one sequence of Q1-Q4. Two sequences of Q1 through Q4 are required minimum for execution of an instruction.





**Fig 3.9 Clock/Instruction Cycle of PIC18F458**

Source: Microchip Document: PIC18FXX8 Data Sheet - DS41159E

Program counter is incremented during every Q1 phase and an instruction is fetched from program memory and transfer to Instruction register is activated during the Q4 phase. Later during the following Q1 through Q4 phase the instruction is decoded and executed. The instruction code is stored in Instruction register during Q1 phase. Decoding and execution of instruction takes place during Q2(Data memory read operation), Q3 and Q4 (data memory destination write operation).

### 3.7 Clock Configuration

A microcontroller's CPU core is operated at a specific frequency at any given point of time. Peripherals and other components are operated at the same or lower frequencies by using frequency scaling circuits. Devices requiring higher performance make use of high clock rates, and lower clock rates are preferred for lower power consumption. Additionally, clock stability, time required to produce a stable clock etc are also important parameters in clock selection. To provide flexibility, configuration features are provided to select among multiple clock sources with different types of clock generators considering efficiency, performance and power requirements.

The PIC18F458 is designed to operate in any one of eight oscillator modes at a given time. Three programmable configuration bits (FOSC2, FOSC1 and FOSC0) are provided to select the oscillator mode. These bits are provided for configuration through the CONFIG1H register presented here. The eight possible modes are presented in Table 3.8. A detailed view of clock types and configurations is presented in clock section.

#### CONFIG1H – Configuration 1 High Register Format

| U-0   | U-0 | R/P-1                       | U-0 | U-0 | R/P-1 | R/P-1 | R/P-1 |
|-------|-----|-----------------------------|-----|-----|-------|-------|-------|
| —     | —   | OSCS $\overline{\text{EN}}$ | —   | —   | FOSC2 | FOSC1 | FOSC0 |
| bit 7 |     |                             |     |     |       |       | bit 0 |

| Bit No  | Bit           | Description                                                                                                                                                                                                                                                                                                                                                     |
|---------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bit 7-6 | Unimplemented | Read as '0'                                                                                                                                                                                                                                                                                                                                                     |
| bit 5   | <u>OSCSEN</u> | Oscillator System Clock Switch Enable bit<br>1 = Oscillator system clock switch option is disabled (main oscillator is source)<br>0 = Oscillator system clock switch option is enabled (oscillator switching is enabled)                                                                                                                                        |
| bit 4-3 | Unimplemented | Read as '0'                                                                                                                                                                                                                                                                                                                                                     |
| bit 2-0 | FOSC2:FOSC0   | Oscillator Selection bits<br>111 = RC oscillator w/OSC2 configured as RA6<br>110 = HS oscillator with PLL enabled/clock frequency = (4 x FOSC)<br>101 = EC oscillator w/OSC2 configured as RA6<br>100 = EC oscillator w/OSC2 configured as divide-by-4 clock output<br>011 = RC oscillator<br>010 = HS oscillator<br>001 = XT oscillator<br>000 = LP oscillator |

**Table 3.8 Oscillator Types**

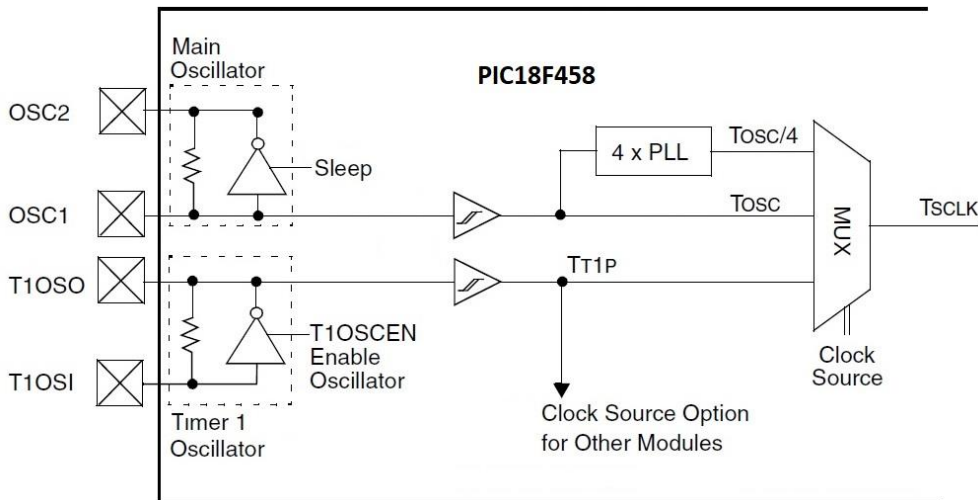
| Mode | Mode Name                                             |
|------|-------------------------------------------------------|
| 1    | LP Low-Power Crystal                                  |
| 2    | XT Crystal/Resonator                                  |
| 3    | HS High-Speed Crystal/Resonator                       |
| 4    | HS4 High-Speed Crystal/Resonator with PLL enabled     |
| 5    | RC External Resistor/Capacitor                        |
| 6    | RCIO External Resistor/Capacitor with I/O pin enabled |
| 7    | EC External Clock                                     |
| 8    | ECIO External Clock with I/O pin enabled              |

The block diagram of the PIC18F458 system clock sources is shown in fig 3.10. The PIC18F458 incorporates two possible clock source pins.

- Main Oscillator pins
- Timer 1 Oscillator pins

The main clock oscillator is connected to the Oscillator pins. The alternative low frequency clock source (typically a 32 kHz crystal) is connected to Timer 1 Oscillator pins. The system clock may be sourced from either of these two clock sources. A switching feature allows to switch between these two clock inputs i.e. switch from the main oscillator to an alternate low-frequency clock source or vice versa. The SCS - System Clock Switch bit in

OSCCON - Oscillator Control register is used for the clock switching purpose. The OSCCON register is programmable, and the format of this register is given below.



**Fig 3.10 Block diagram of system clock sources**

Source: Microchip Document: PIC18FXX8 Data Sheet - DS41159E

**OSCCON - Oscillator Control Register Format**

|       |     |     |     |     |     |     |       |
|-------|-----|-----|-----|-----|-----|-----|-------|
| U-0   | U-0 | U-0 | U-0 | U-0 | U-0 | U-0 | R/W-1 |
| —     | —   | —   | —   | —   | —   | —   | SCS   |
| bit 7 |     |     |     |     |     |     | bit 0 |

| Bit No  | Bit           | Description                                                                                                                                                                                                                                                    |
|---------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| bit 7-1 | Unimplemented | Read as '0'                                                                                                                                                                                                                                                    |
| bit 0   | SCS           | System Clock Switch bit<br>When <u>OSCS</u> configuration bit = 0 and T1OSCEN bit is set:<br>1 = Switch to Timer1 oscillator/clock pin<br>0 = Use primary oscillator/clock input pin<br>When <u>OSCS</u> is clear or T1OSCEN is clear:<br>Bit is forced clear. |

Setting the Oscillator Switching Enable (OSCS) bit in CONFIG1H Configuration register to 0 enables the clock switching feature. The SCS - system clock source switching bit in OSCCON register when set to 0, selects the main oscillator to be the system clock source and the PIC18F458 operates in the mode based on the FOSC2:FOSC0 configuration bits. The SCS bit when set to 1, selects the Timer 1 oscillator to be the system clock source.

To make the Timer1 oscillator as system clock source, the Oscillator System Clock Switch Enable bit OSCS in CONFIG1H register needs to be cleared and Timer1 Oscillator Enable bit T1OSCEN in T1CON needs to be set. If Timer1 oscillator is not enabled or Oscillator System Clock Switch Enable bit OSCS is set (switch disabled), the system clock source

switching bit is forced clear, i.e. the system clock is sourced from main oscillator. The SCS bit is cleared on Reset. When the timer1 oscillator is selected as a system clock source, the PIC18F458 can execute in a Low-Power mode.